

Correlation of meltpool characteristics and residual stresses at high laser intensity for metal lpbf process

Alexandre Staub, Adriaan B. Spierings & Konrad Wegener

To cite this article: Alexandre Staub, Adriaan B. Spierings & Konrad Wegener (2018): Correlation of meltpool characteristics and residual stresses at high laser intensity for metal lpbf process, *Advances in Materials and Processing Technologies*, DOI: [10.1080/2374068X.2018.1535643](https://doi.org/10.1080/2374068X.2018.1535643)

To link to this article: <https://doi.org/10.1080/2374068X.2018.1535643>



Published online: 20 Oct 2018.



Submit your article to this journal [↗](#)



Article views: 6



View Crossmark data [↗](#)



Correlation of meltpool characteristics and residual stresses at high laser intensity for metal lpbf process

Alexandre Staub ^a, Adriaan B. Spierings^a and Konrad Wegener^b

^aInspire, Innovation Center for Additive Manufacturing Switzerland (icams), St. Gallen, Switzerland; ^bETH Zurich, Institute of Machine Tools and Manufacturing, Zurich, Switzerland

ABSTRACT

Selective Laser Melting (SLM) commonly referred as Metal Laser Powder Bed Fusion (LPBF) processes, proved in the last decade to be suitable for the manufacturing of complex metallic components. In order to fulfil industrial needs from various industries (e.g. aerospace, tooling, energy production, medical), Machine manufacturers have increased the productivity of the LPBF-process mainly by increasing the number and maximum power of lasers. However, this strategy also affects the magnitude of residual stresses generated in the consolidated material. This study analyses the residual stresses in SS316L components by XRD measurement, where a correlation to the meltpool dimensions could be found and verified for different laser power and scan speeds. The results provide fundamentals to assess the gain in productivity, and to establish generic guidelines for the optimization of residual stresses at high laser power.

ARTICLE HISTORY

Accepted 10 October 2018

KEYWORDS

Additive manufacturing; laser powder bed fusion (LPBF); residual stresses; X-Ray Diffraction (XRD); meltpool; Stainless Steel 316L

1. Introduction

Selective Laser Melting (SLM) (PBF-LB/M according to ISO/ASTM DIS 52,900:2018), commonly Laser Powder Bed Fusion (LPBF) is one of the most promising Additive Manufacturing (AM) technologies for the manufacturing of metallic three-dimensional structures having typical sizes from tenths of millimetres to tenths of centimetres [1]. Bourell *et al.* [2] presented a review of the different AM-processes and highlighted the bottlenecks such as remaining porosity, balling phenomena, surface finishing, cracks, and chemical degradation. Schmidt *et al.* [3] also discussed the opportunities of the LPBF process to become a serial production process and its limitations in terms of machines design, data preparation and material quality. Notably the residual stresses remaining in the parts are identified by Patterson *et al.* [4] as a bottleneck for further technology development. Indeed, the high stress levels in the parts can lead to build job interruption in case of in-process deformation, or to critical part distortions once they are removed from the build plate. In fact, no prediction of the final part accuracy is possible without the knowledge of the residual stress state, which is greatly influenced by the part geometry and the build-up strategy. In addition, the residual stresses limit their industrial performance during use.

Independently from the manufacturing technique, all metallic materials are subject to residual stresses either coming from mechanical or thermal solicitation during the manufacturing process. Complex residual stress states in LPBF originate from the high temperature gradient induced by the high intensity laser process, and according to Kruth *et al.* [5] strongly depends on the Marangoni convection. Indeed, the extremely fast temperature change in the process is leading to two phenomena. Both the temperature gradient mechanism (TGM) during the heating phase and a thermal shrinkage of the solidified melt-pool during the cooling phase yield to the formation of high residual stresses as a result of surrounding solid material constraining the solidification of the molten material [6].

A lot of research has been performed to characterise residual stresses induced by the LPBF process. The different mechanisms of residual stresses formation are having different scales, and are classified in three types as proposed by Withers *et al.* [7], being the atomic level (type III), the grain level (type II) and the macroscopic level (type I). For each stress type specific tools and methods for quantification are used, which do not permit the quantification of the other types. The residual stresses studied in most of the literature about the LPBF process are from type I, being macro-scale stresses, as these stresses are the most relevant in terms of part deformation. Measurement methods such as geometrical deformation permit a qualitative assessment of the residual stresses magnitude using a beam curvature after removal of the support structure binding to the build plate, as used by Safronov *et al.* [8], while Ghasri-Khouzani *et al.* [9] used the overall deformation of disk artefacts for the same purpose. However, the use of neutron diffraction [9,10] or the hole drilling method [11] permit a better understanding of the residual stress profiles over the depth of a sample. Casavola *et al.* [11] identified that due to the partial re-melting of the layers through the build, residual stresses would be reduced in the depth of the sample. Bartlett *et al.* [12] proposed a new methodology to *in-situ* monitor residual stresses during the LPBF-process, by using digital image correlation. These results permit to establish stress maps for simple geometries. The authors also highlighted the high dependencies of the stress state on the geometry of the sample. X Ray Diffraction (XRD) measurements have been performed to assess the stress state in a surface near region, hence within the last top-layer of the sample. Vrancken *et al.* [13] used XRD to identify that the main stress direction is in the scan direction, while Mercelis *et al.* [6] applied chemical etching to assess the stress state of layers beneath the top layer, giving information about the stress profile. Numerous analytical studies [14–16] analysed the effects of different scanning strategies on residual stress reduction, concluding that the reduction of the scanning vector length and the application of a chessboard island scan strategy are beneficial for stress reduction.

However, so far not sufficient work has been put into the characterisation of the residual stresses for higher laser intensities up to 5.8 kW/mm^2 , and the dependency of residual stresses from the properties of the melt-pool created by the laser beam. Hence, this paper intends to advance the state of the art in the comprehension of these phenomena. By the characterisation of the residual stresses and the correlation with the meltpool dimensions, this paper aims at providing a comparison basis for the understanding of the residual stresses in LPBF at various laser powers using a constant

methodology in term of manufacturing by keeping all slicing parameters constant through the increase of the laser power.

2. Materials and methods

2.1. Sample definition and LPBF parameters

10x10x10mm³ cube samples were produced in SS316L, as a widely adopted material in metal additive manufacturing. A ConceptLaser M2 machine, equipped with a SPI 400W Nd-YAG laser source operated in continuous wave mode, was used. The laser spot diameter in the processing plane was 130 µm. The slicing thickness (L) and hatching distance (h) were respectively of 40 µm and 90 µm. The classic meander scanning strategy with a 90° rotation applied between each layer was chosen as it has been shown to be the most promising strategy for high laser intensities [17].

Prior to the analysis of meltpool characteristics and residual stress state, a process window for a material density of $\rho \geq 99.5\%$ was developed for each laser power level (P). By the variation of the scanning speed (v) samples were produced and measured for $P = 100 \text{ W}, 150 \text{ W}, 200 \text{ W}, 250 \text{ W}, 300 \text{ W}, \text{ and } 350 \text{ W}$. The common characteristics to all process windows is that they use the same volumetric energy density range, e_d , as defined in Equation (1).

$$e_d = \frac{P}{L * h * v} \text{ (J.mm}^{-3}\text{)} \quad (1)$$

2.2. Analysis

The material density analysis was performed with a non-destructive density measurement methodology using Archimedes' principle applied in acetone. As pointed out by Spierings *et al.* [18] this analysis is easy and yields to quick and reliable results in comparison to other analysis methods, e.g. X-ray neutron imaging, or optical micrograph analysis.

The meltpool characteristics were analysed by a cutting, embedding, polishing and etching procedure in the transversal plane. After ODS (oxide dispersion solution) polishing down to $R_a = 50 \text{ nm}$ the samples were etched for 30 s using V2A etchant (Schmitz-Metallographie, Germany). The analysis was performed using an optical light microscope Leica DM6 at various magnifications.

Three different samples per laser power level were selected for the residual stress state quantification. For comparison purposes, the three samples will each have respectively the same energy density for each laser power. The samples are produced using energy densities, e_d , of 43, 62 and 79 J/mm³, and having all a material density $\rho \geq 99\%$ (see arrows on Figure 1).

The residual stresses were measured using an X Ray Diffractometer Stresstech Xstress 3000 G3 using Mn radiation at wavelength of 0.2103 nm, equipped with a 1 mm diameter collimator. The measurements of the samples are obtained for a 2θ value of 152.3°, at nine different tilt angles being: 0°, ± 18.3°, ± 26.4°, ± 33.0°, ± 39.0°. The total exposure

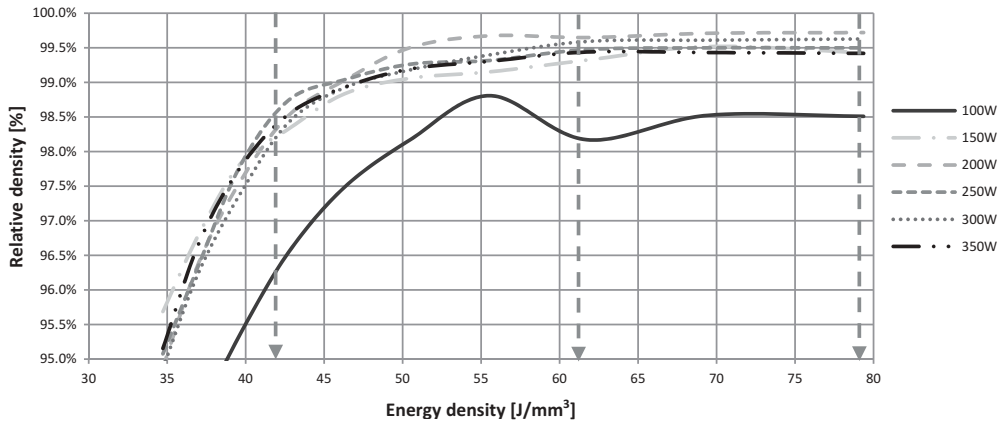


Figure 1. Relative Material density-gain curve along the energy density for several laser powers. Arrows indicating the selected samples for XRD analysis.

time is 65 s. The two principal in-plane directions σ_{11} and σ_{22} are computed and the resulting stress is deduced.

Stress relieve operations were realised at a temperature of 800°C followed by a natural furnace cooling [19].

3. Results and discussion

3.1. Process window development

In order to keep comparable results, the energy density range to establish the processing windows is kept the same for all laser powers. Consequently, and knowing the fixed parameter presented above, it is impossible to establish a suitable processing window for a laser power of 100W, as shown on Figure 1. This is due to the meltpool size being not large enough to correctly connect to the neighbouring scan tracks, resulting in connection defects and correspondingly a high material porosity. Indeed, it has been shown in previous studies that such energy density range could easily process SS316L, but only with other process parameters, notably a smaller hatch distance and layer thickness.

All the other five processing windows developed for $P = 150\text{ W}$ to 350 W successfully reached the expected material density level and the above-described samples were used for residual stresses measurement.

3.2. Meltpool characteristics

The meltpool dimensions were measured in the cross section of the samples, averaged and represented using elliptical shapes, representing all typical meltpool geometries, as in the present data set there were no ‘bell shaped’ meltpool observed due to the selected manufacturing parameters. The aspect ratio (AR), being the ratio between the width and the depth of the sample, was calculated to highlight the meltpool shape. An $AR > 1$ represents shallow and flat meltpools, while $AR < 1$ represents deep and thin meltpools. The characteristics of the meltpools are summarized in Table 1.

Table 1. Meltpools characteristics in the different processing conditions.

P [W]	Ed [J/mm ³]	AR	Area [μm ²]
150	43	1.53	10,033
	62	1.51	19,554
	79	1.64	22,616
200	43	1.15	13,344
	62	1.13	24,951
	79	1.30	24,362
250	43	1.28	16,496
	62	0.95	22,425
	79	1.19	46,863
300	43	1.28	23,164
	62	0.81	20,793
	79	0.71	29,863
350	43	1.10	24,075
	62	0.67	28,612
	79	0.64	34,508

As shown in Figure 2, meltpool shapes tend to elongate in depth with increasing laser power assuming a constant energy density. Typical AR values are between 1.51 to 0.67 for an energy density, e_d , of 62 J/mm³. For each laser power P in the power range [150 W; 250 W], the AR tends to increase with the corresponding energy density increase, in the domain where the consolidated material is considered dense ($\rho \geq 99\%$). While for higher laser power P > 250 W the contrary effect is observed. Therefore, the assumption can be made that this change in the meltpool shape is linked to a change in the dynamic behaviour of the meltpool during the liquid phase and its energy dissipation to the surrounding material, thus influencing the residual stress state. Once a threshold of P = 250W is exceeded, and for energy densities permitting to produce dense material, the welding mode switches from convection to keyhole welding, as discussed by King *et al.* [20] and Staub *et al* [17]., resulting in more elongated and deeper meltpool shapes.

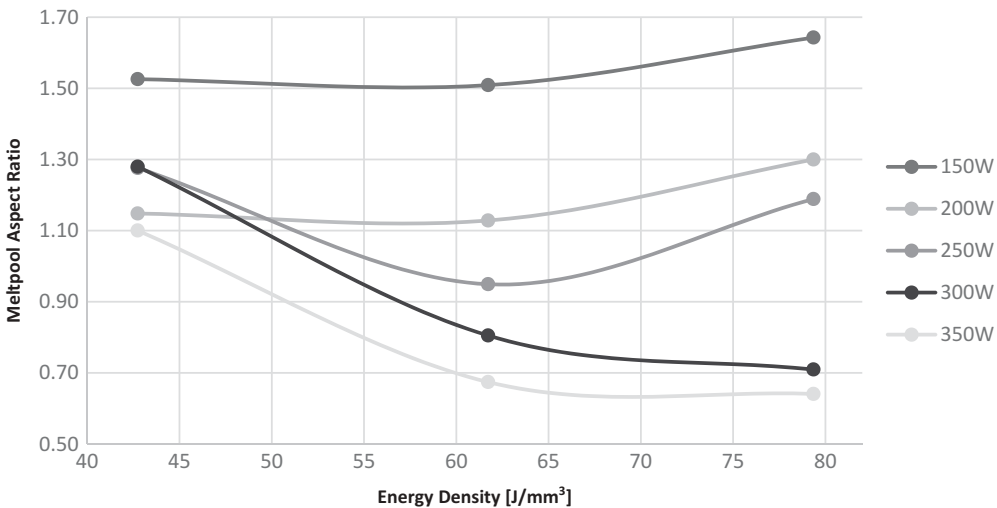


Figure 2. Meltpool characteristics at different laser powers and energy densities.

The increase in the energy brought to the powder layer, considering a constant laser power, will also increase the area of the meltpool cross-section as shown in Table 1. Indeed, the time given to the growth of the meltpool per unit of length is longer for a higher energy density due to the slower scanning speed. This volume increase goes along with a change in the aspect ratio AR , showing the tendency of the meltpool not to increase its width proportionally to the depth. This anisotropic growth of the meltpool size is related to the given laser spot diameter that is comparable or slightly smaller than the meltpool width, thus explaining the tendency of the meltpool to switch to the keyhole mode.

3.3. Residual stresses

Figure 3 shows the residual stresses in dependence of the aspect ratio. A steady increase of the residual stress state can be observed for increasing AR . A flatter meltpool induces higher residual stresses on the surface due to its larger width in comparison to a deeper and thinner meltpool. This leads to the conclusion that deeper meltpools are beneficial for lower in-plane residual stresses. However, this measurement does not give any insights on the residual stresses in the build direction, which might be increased by the meltpool elongation.

At a laser power of $P = 250$ W the residual stresses are stable through the increase of the energy input and in the magnitude of the yield stress of the bulk material. This behaviour has also to be correlated with the meltpool characteristics, which show relatively large but stable meltpools for this parameter set.

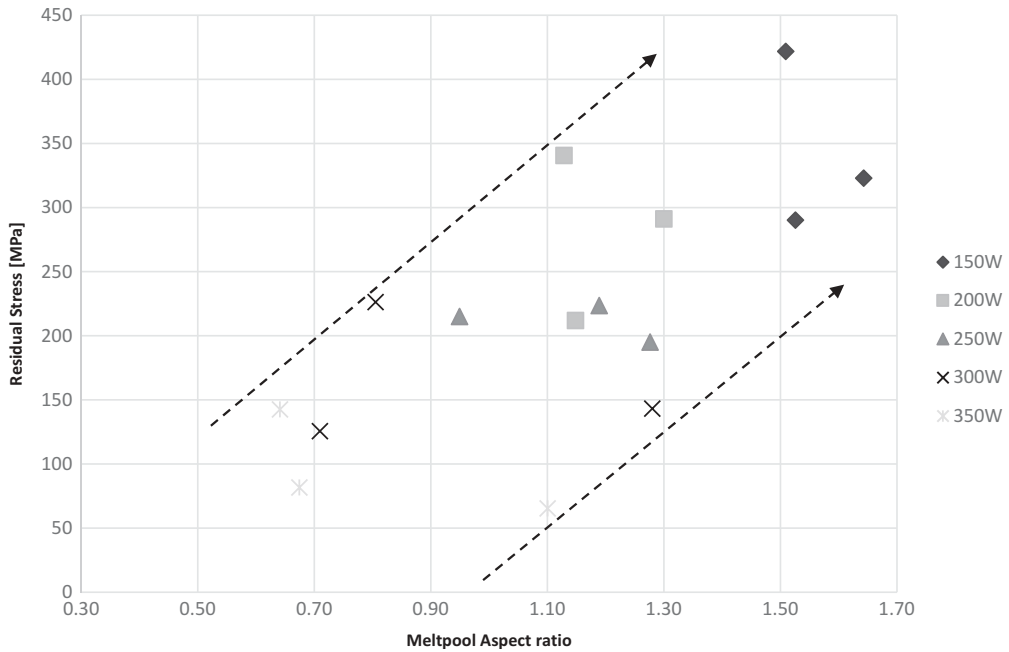


Figure 3. Residual stresses measured at different laser powers in dependence of the melt-pool aspect ratio.

For the other laser powers, except for $P = 350 \text{ W}$ were the stresses increase with the energy input, e_d , the residual stress state shows the same behaviour. The samples having a lower material density result in lower residual stresses, due to a different mechanical behaviour arising from the higher pore concentration. The residual stresses increase for samples having a relative material density $\rho \geq 99.5\%$ and decrease again for even higher energy inputs. This effect arise from the increase of the sample temperature at higher energy density, thus increasing the ductility of the material and lowering the resulting residual stresses.

Removing approximately $-150 \text{ }\mu\text{m}$ by etching the sample produced using 250 W with an energy density 62 J/mm^3 , hence removing the last molten layer leaves a zone that was not remolten by the last scan tracks. This allows the measurement of residual stresses within the material, yielding to a resulting stress of 593 MPa at a depth of the meltpool bottom. This drastic increase of the stresses compared to measurements done at a surface-near region highlights the complexity of the stress distribution in LPBF-produced samples. The highest temperature of the meltpool is found at its bottom according to Zhang *et al.* [21]. The even steeper temperature gradient in the neighbourhood of the meltpool bottom explains this drastic increase in residual stresses.

The stress-relieved samples, produced at 300W , showed small residual stresses in the magnitude of 50MPa , proving the reliability of the measurement for this study.

4. Conclusion

This study investigates the meltpool size and shape in dependence of the laser power at comparable processing conditions (i.e. constant scanning strategies and energy input range) and correlates them with the meltpool shape and area. Several effects are identified:

- (1) The shape and area, of the meltpool correlate with the residual stresses on the top layer of the sample. Thereby, larger and flatter meltpools induce higher in-plane stresses in both directions.
- (2) The higher laser intensities influence single-track residual stresses. Nevertheless, the overall increase of the temperature of the sample appears to be beneficial for the relief of stresses. For this reason, for a proper comparison of stress measurement data, the temperature of the samples needs to be considered, which is highly dependent of the volume, geometry and orientation of the part.
- (3) The complexity of the residual stresses also needs to be investigated in the build direction in order to draw optimisation guidelines for the LPBF process.

The results of this study tend to favour the keyhole melting mode as beneficial for lower residual stresses. In addition, detrimental effects of the keyhole welding mode, such as e.g. the formation of keyhole porosity, and typical complex features on real AM-parts, e.g. fine lattice structures, and the quality of overhanging surfaces, which are more complex to manufacture at high laser intensities should be considered. This study serves as a basis for further development of simulation models taking into account the meltpool characteristics. The results are also a contribution to the general optimisation of the LPBF process. Nevertheless, a visual inspection permits to realise that the residual stress state needs to

be compared to other dimensions in order to optimise the process. Residual stress reduction cannot be the only goal to improve the LPBF process. Further work to correlate different physical dimension such as residual stresses in build direction, surface quality of overhanging surface, mechanical behaviour are necessary to lead the optimisation of the process.

Acknowledgments

The authors would like to kindly acknowledge the company Stresstech GmbH for leading fast and effective XRD measurements of the samples.

Disclosure statement

No potential conflict of interest was reported by the authors.

ORCID

Alexandre Staub  <http://orcid.org/0000-0001-7481-1939>

References

- [1] Herzog D, Seyda V, Wycisk E, et al. Additive manufacturing of metals. *Acta Materialia*. Vol. 117. 2016. p. 371–392.
- [2] Bourell D, Kruth JP, LEu M et al. Materials for additive manufacturing. *CIRP Annals - Manufacturing Technology*. 2017;66(2):659–681.
- [3] Schmidt M, Merklein M, Bourell D Laser based additive manufacturing in industry and academia. *CIRP Ann*. 2017;66(2):561–583.
- [4] Patterson AE, Messimer SL, Farrington PA. Overhanging Features and the SLM/DMLS Residual Stresses Problem: review and Future Research Need. *Technologies*. 2017;5(2).
- [5] Kruth JP, Froyen L, Van Vaerenbergh J, et al. Selective laser melting of iron-based powder. *J Of Mater Process Technol*. 2004 Jun 10;149(1–3):616–622.
- [6] Mercelis P, Kruth JP. Residual stresses in selective laser sintering and selective laser melting. *Rapid Prototyping Journal*. 2006;12(5):254–265.
- [7] Withers PJ, Bhadeshia HKDH. Residual stress part 2 - Nature and origins. *Mater Sci Technol*. 2001;17(4):366–375.
- [8] Safronov VA, Khmyrov RS, Kotoban DV, et al. Distortions and residual stresses at layer-by-layer additive manufacturing by fusion. *J Manufacturing Sci Engineering, Trans ASME*. 2017;139(3). Art. No. 031017.
- [9] Ghasri-Khouzani M, Peng H, Rogge R et al. Experimental measurement of residual stress and distortion in additively manufactured stainless steel components with various dimensions. *Mater Sci Eng*. 2017;707:689–700.
- [10] Brown DW, Bernardin JD, Carpenter JS, et al. Neutron diffraction measurements of residual stress in additively manufactured stainless steel. *Materials Science and Engineering A*. Vol. 678. 2016. p. 291–298.
- [11] Casavola C, Campanelli SL, Pappalettere C. Preliminary investigation on distribution of residual stress generated by the selective laser melting process. *J Strain Anal Eng Des*. 2009;44(1):93–104.
- [12] Bartlett JL, Croom BP, Burdick J, et al. Revealing mechanisms of residual stress development in additive manufacturing via digital image correlation. *Additive Manufacturing*. 2018;22:1–12.

- [13] Vrancken B, Wauthle R, Kruth JP, et al., “Study of the influence of material properties on residual stress in selective laser melting,” in *24th International SFF Symposium - An Additive Manufacturing Conference, Austin, SFF 2013*, 2013, pp. 393–407.
- [14] Dai K, Shaw L. Distortion minimization of laser-processed components through control of laser scanning patterns. *Rapid Prototyping Journal*. 2002;8(5):270–276.
- [15] Cheng B, Shrestha S, Chou K. Stress and deformation evaluations of scanning strategy effect in selective laser melting. *Additive Manufacturing*. 2016;12:240–251.
- [16] Liu Y, Yang Y, Wang D. A study on the residual stress during selective laser melting (SLM) of metallic powder. *Int J Advanced Manufacturing Technol*. 2016;22:1–10.
- [17] Staub A, Spierings AB, Wegener K, “Selective Laser Melting at High Laser Intensity: overhang Surface Characterization and Optimization “ in *Direct Digital Manufacturing Conference*, Berlin, 2018, ISBN 978-3-8396-1320-7.
- [18] Spierings AB, Schneider M, Eggenberger R. Comparison of density measurement techniques for additive manufactured metallic parts. *Rapid Prototyping Journal*. 2011;17(5):380–386.
- [19] Kurzynowski T, Gruber K, Stopyra W, et al. Correlation between process parameters, microstructure and properties of 316 L stainless steel processed by selective laser melting. *Mater Sci Eng*. 2018;718:64–73.
- [20] King W E, Barth H D, Castillo V M et al. Observation of keyhole-mode laser melting in laser powder-bed fusion additive manufacturing. *J Mater Process Technol*. 2014;214 (12):2915–2925.
- [21] Zhang D, Zhang P, Liu Z, et al. Thermofluid field of molten pool and its effects during selective laser melting (SLM) of Inconel 718 alloy. *Additive Manufacturing*. 2018;21:567–578.